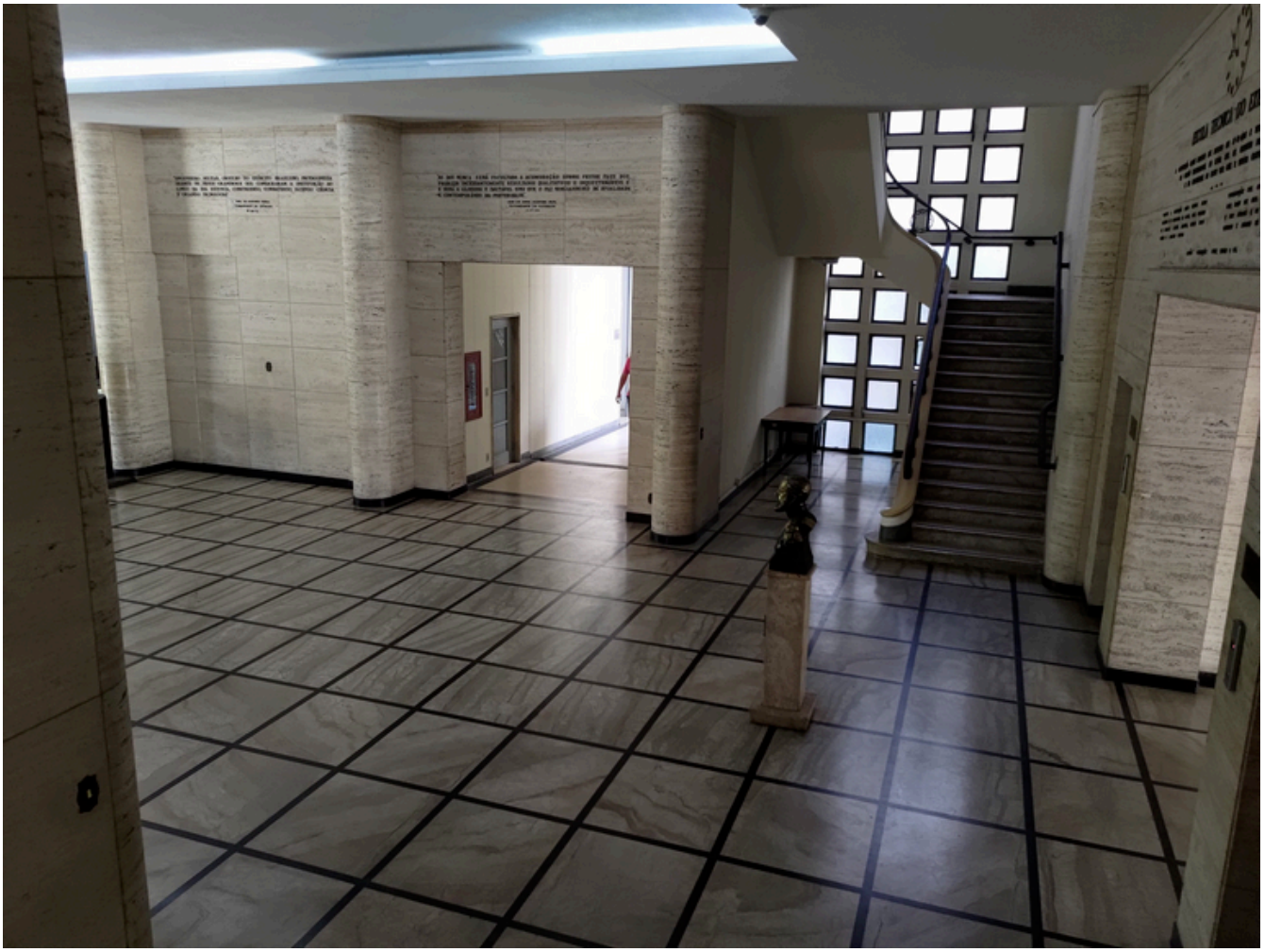
Problem B. Don't Get Caught

Time Limit	2000 ms
Mem Limit	262144 kB
OS	Windows

Today is a beautiful day in Rio de Janeiro and all of the students at `IME` (known as the `Institute of Missions that are Enthralling`) decide to go enjoy the beach that's right in front of the institute, all except for `Kauã`... `Kauã` needs to stay at `IME` to complete his mission of safeguarding the university. Although he's not allowed to leave, he decides he wants to sneak out to have fun with his friends, but obviously he doesn't want to get caught! You sympathize with this situation and decide to help him develop an escape plan.

`Kauã` reaches the main entrance hall and is greeted with this large hallway containing a number of guards watching the place. Since it's quite the boring night and they think that the institute is safe, the guards remain static throughout the entire night, looking solely in one direction. After carefully analyzing their positions, `Kauã` hands to you a map of the main entrance hall. It consists of a grid with dimensions $n \times m$, in which Kauã starts at position $(1, 1)$ and has to get to the main entrance itself, at (n, m). The guards will be situated in other positions of the grid, however, due to some restrictions, there's never someone at $(1, 1)$ nor at (n, m).

`Kauã` asks you to help him determine whether or not he's able to traverse the main entrance hall. If he's able to, he asks you to show in the map what's the smallest path he's able to take.



The picture depicts where Kauã needs to traverse.

Input

The first line of the input consists of two integers n, m $(1 \leq n, m \leq 2 \times 10^3)$ — the dimensions of the main entrance hall.

The next n lines will contain a string of m characters $(a_{i,1}, a_{i,2}, \ldots, a_{i,m})$, indicating the layout of the ith line.

$a_{i,j}$ can be either `.`, >, v, <, ^ — indicating what's present on that position.

- If $a_{i,j} =$ `.`, then the position is empty.
- If $a_{i,j} =$ >, v, <, ^, then the position contains a guard that is facing east, south, west, and north respectively. The guard's line of sight is one square wide and extends straight in the direction the person is facing, blocked only by another person.

Output

If it is impossible to reach the main entrance itself, then print out -1. Otherwise, print out the grid once more, substituting the positions Kauã is supposed to pass through with an X. This should be the shortest path, and if there are multiple shortest paths, you can print out any of them.

Examples

Input	Output
5 5 ..... .>.v. ..... .^.<.	X.... X>.v. X.... X^.<. XXXXX

Input	Output
3 6 .v.... .^.v.. ...^..	XvXXX. X^XvX. XXX^XX

Input	Output
4 4 .... ..^. .<..	-1